MARK SCHEME · TEACHER COPY

10BR · Chapter 9: Databases

Cambridge IGCSE Computer Science 0478

Paper 2 Algorithms, Programming and Logic

Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Total: 75 marks

Question 1 · Feb/March 2023 Paper 22, original Q9 (a),(b),(c),(d),(e) [9 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)</td><td>20</td><td>1</td></tr>
<tr><td>1(b)(i)</td><td>CatNo</td><td>1</td></tr>
<tr><td>1(b)(ii)</td><td>It is a unique identifier // no repeated values</td><td>1</td></tr>
<tr><td>1(c)</td><td>Two marks for 4 correct data types or one mark for 2 or 3 correct data types
<table>
<tr><th>Field</th><th>Data type</th></tr>
<tr><td>CatNo</td><td>Text/Alphanumeric</td></tr>
<tr><td>Title</td><td>Text/alphanumeric</td></tr>
<tr><td>Fiction</td><td>Boolean</td></tr>
<tr><td>Price</td><td>Real</td></tr>
</table></td><td>2</td></tr>
<tr><td>1(d)</td><td>One mark for each correct row
BK08 The Princesses' Story B Penn
BK31 Networking for Beginners A Smith</td><td>2</td></tr>
<tr><td>1(e)</td><td>One mark if two correct or two marks if completely correct
Title
BookList
Author = "B Penn" // Author = 'B Penn' // Author Like "B Penn"</td><td>2</td></tr>
</table>

Question 2 · May/June 2023 Paper 21, original Q10 (a),(b),(c) [8 marks]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(a)</td><td>One mark for the correct field name
One mark for the correct reason

For example:

TVCode
Each entry in this field is a unique identifier</td><td>2</td></tr>
</table>

Total: 75 marks

2(b)	**Two** marks for four correct answers. **One** mark for two or three correct answers. **Field** / **Data type** TVCode — Text ScreenSize — Integer SmartTV — Boolean Price$ — Real	**2**

2(c)	**One** mark for each correct answer
	
	ScreenSize
	Price$
	FROM
	YES
	
	Correct code:
	
	`SELECT TVCode, ScreenSize, Price$`
	`FROM TVRange`
	`WHERE SmartTV = YES;`
		4

Question 3 · Oct/Nov 2023 Paper 21, original Q9 (a),(b),(c) [10 marks]

Question	Answer	Marks
3(a)	**One** mark for each appropriate field name **and** correct data type CatNo — text/alphanumeric Description — text/alphanumeric StockLevel — integer Price — real Painted — Boolean	**5**
3(b)(i)	CatNo	**1**
3(b)(ii) i)	Unique identifier	**1**

3(c)	`SELECT` lists the fields to be displayed `FROM` identifies the table `WHERE` identifies the search criteria	3

Question 4 · May/June 2024 Paper 23, original Q8 (a),(b),(c) [10 marks]

Question	Answer	Marks
4(a)	Fields – 6 Records – 11	2

4(b)	MP1 Correct models of planes selected
	MP2 Correct Years of planes selected
	MP3 Correct numbers of engines selected with data all matching for each record
	MP4 Data sorted correctly with no additional fields or punctuation
	
	Correct output:
	
	`314 Clipper 1936 4`
	`C-47 Dakota 1942 2`
	`Nimrod 1966 4`
	`DC-10 1970 3`
	`Concorde 1973 4`
		4

4(c)	MP1 Correct additional keywords used – `FROM`, `WHERE`
	MP2 Correct `SELECT` field – `Model` with no additional fields
	MP3 Correct table name - `Hangar1`
	MP4 Correct search criteria - `Airworthy = Y // Airworthy`
	
	Correct code:
	
	`SELECT ID, Model`
	`FROM Hangar1`
	`WHERE Airworthy = Y;// Airworthy;`
		4

Total: 75 marks

Question 5 · Oct/Nov 2024 Paper 23, original Q4 [6 marks]

Question	Answer	Marks
5	**One** mark for each correct term Database tables consist of **rows/columns/fields/records** and **columns/rows/fields/records** Rows are **records**. **columns** are fields. Structured Query Language (SQL) **scripts** are used to query data. A **primary key** uniquely identifies a record.	6

Question 6 · May/June 2025 Paper 21, original Q8 (a),(b),(c) [7 marks]

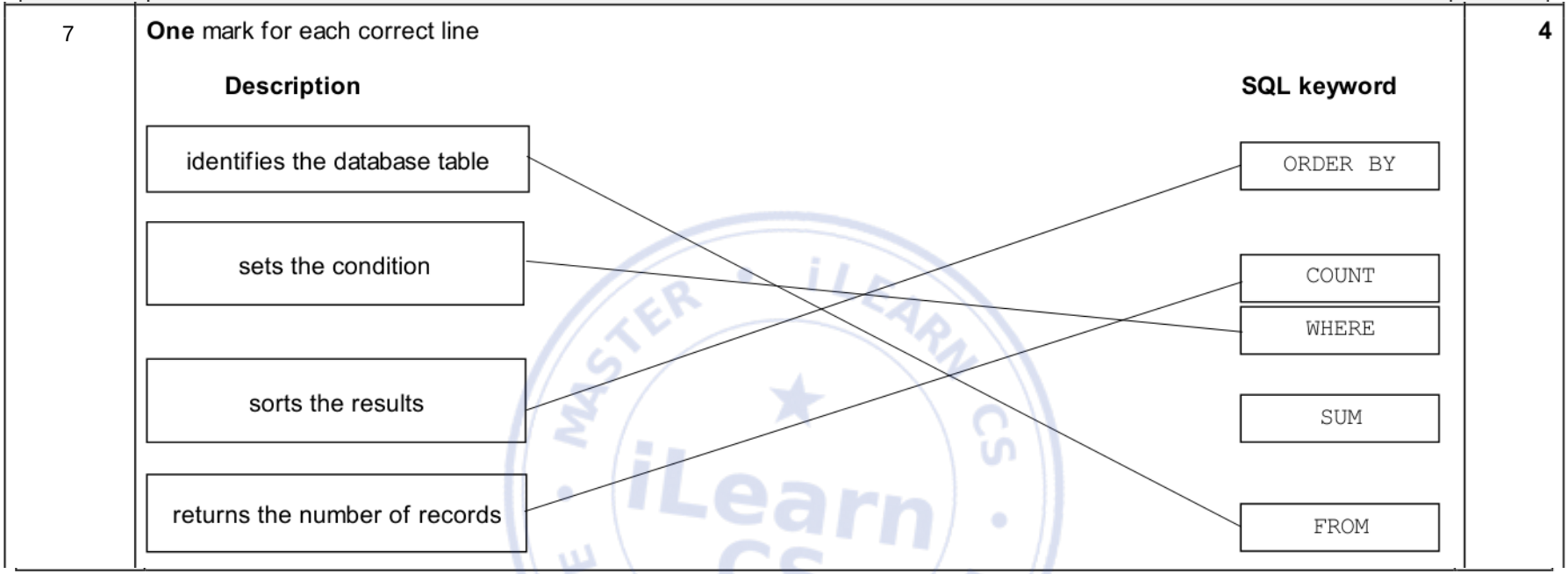
Question	Answer	Marks
6(a)(i)	Boolean	1
6(a)(ii)	There are **only** 2 responses / **Only** Yes and No / True and False	1
6(b)	**Two** marks for all three correct **One** mark for one/two correct BC02 The Fountain 9:10 BC08 Sunset Quay 12:01 BC27 The University 16:45	2

6(c)	**One** mark per correct statement
	
	`SELECT Destination`
	`FROM BUS`
	`WHERE Return = Yes;`
		3

Total: 75 marks

Question 7 · Oct/Nov 2025 Paper 21, original Q1 [4 marks]

Question	Answer	Marks
7	**One** mark for each correct line	4



Question 8 · Oct/Nov 2025 Paper 22, original Q7 (a),(b),(c),(d) [10 marks]

Question	Answer	Marks
8(a) i)	It is used to uniquely identify **records**	1

Total: 75 marks

8(b)	**Two** marks for all four fields correctly identified **One** mark for two or three fields correctly identified **Data type** / **Field** Boolean: `CoastalHabitat` Integer: `Lifespan` Real: `Length` // `Wingspan` Text: `Species` // `Family`	**2**

8(c)	**One** mark per mark point, **max three**
	
	MP1 Correct data – spelt correctly
	MP2 Correct layout – three columns, with no additional punctuation
	MP3 Correct order (vertical and horizontal)
	
	Correct output:
	
	`Bewick's swan 120.2 190.5`
	`Whooper swan 150.0 230.0`
	`Mute swan 150.0 220.2`
		3

8(d)	**One** mark per correct line, **max four**
	
	`SELECT Species, Family, Lifespan`
	`FROM WATERFOWL`
	`WHERE Lifespan >= 10 ORDER BY Species;`
		4

Question 9 · May/June 2026 Paper 23, original Q8 (a),(b),(c) [11 marks]

Question	Answer	Marks	Guidance
9(a)	**One** mark per mark point: • primary key: `SubjectCode` • Reason: it is unique for that subject and level	**2**	

9(b)	**One** mark per mark point: • `PracticalExams/TheoryExams/MinimumClassSize/MaximumClassSize/CourseLength/TheoryExams` • must be represented by whole numbers • `CourseWork` • this is a yes/no field, either there is coursework or there is **no** coursework	**4**	

9(c)	**One** mark per mark point:
	
	• At least two correct fields in `SELECT`
	• Remaining correct field in `SELECT`
	• Correct `FROM OPTION_SUBJECTS`
	• Correct `WHERE MinimumClassSize < 10`
	• Correct `ORDER BY SubjectCode;`
	
	Correct code:
	
	`SELECT SubjectCode, SubjectName, Level`
	`FROM OPTION_SUBJECTS`
	`WHERE MinimumClassSize < 10`
	`ORDER BY SubjectCode (ASC);`
		5	Ignore case on field names, but spelling must match names in table.
			
			If space in field/table name, penalise once and FT.
			
			The default ordering is ascending, so award the mark if ASC is there or not. Penalise if anything else foe example if DESC or ALPHABETICAL is included instead.